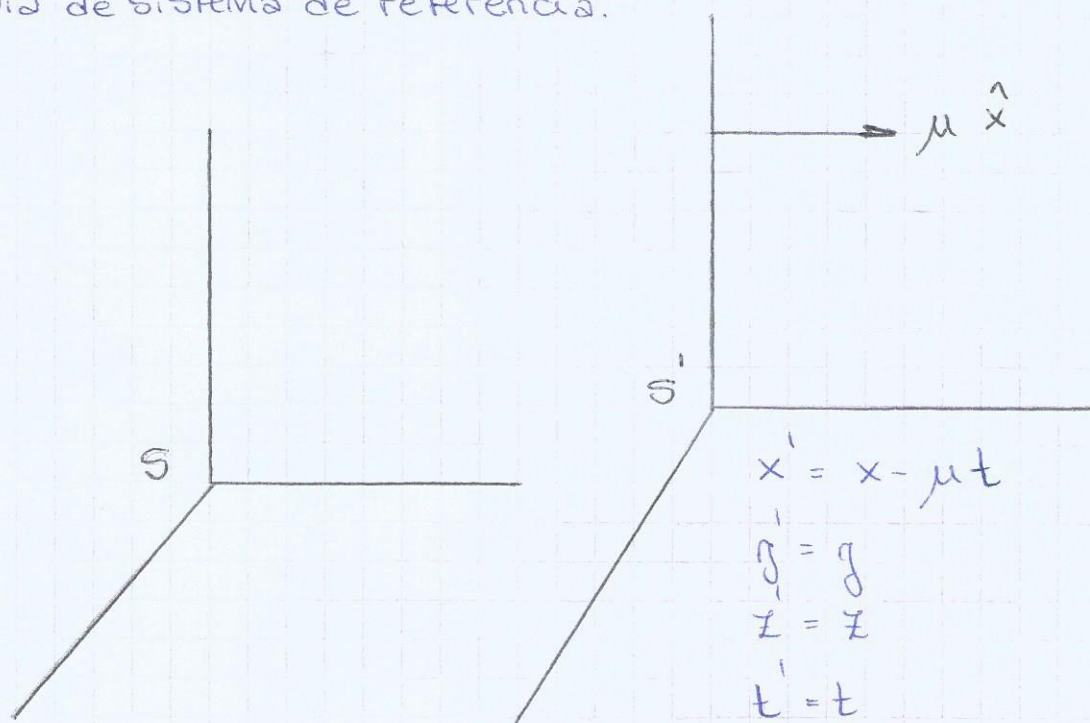


Unidad XIII: Relatividad

- Invariante: Algo que tiene el mismo valor en todo sistema de referencia.
- Covariante: Las leyes tienen la misma forma, cuando cambia de sistema de referencia.



$$dW = dE_c$$

$$dW = F dr$$

$$\begin{aligned} dW &= m \dot{r}' dr' = m \dot{r}' (r') dt \\ &= m \dot{r}' \frac{dr'}{dt} dt = m \dot{r}' dr' \\ &= \frac{1}{2} m \frac{dr'^2}{dt} = dE_c' \end{aligned}$$

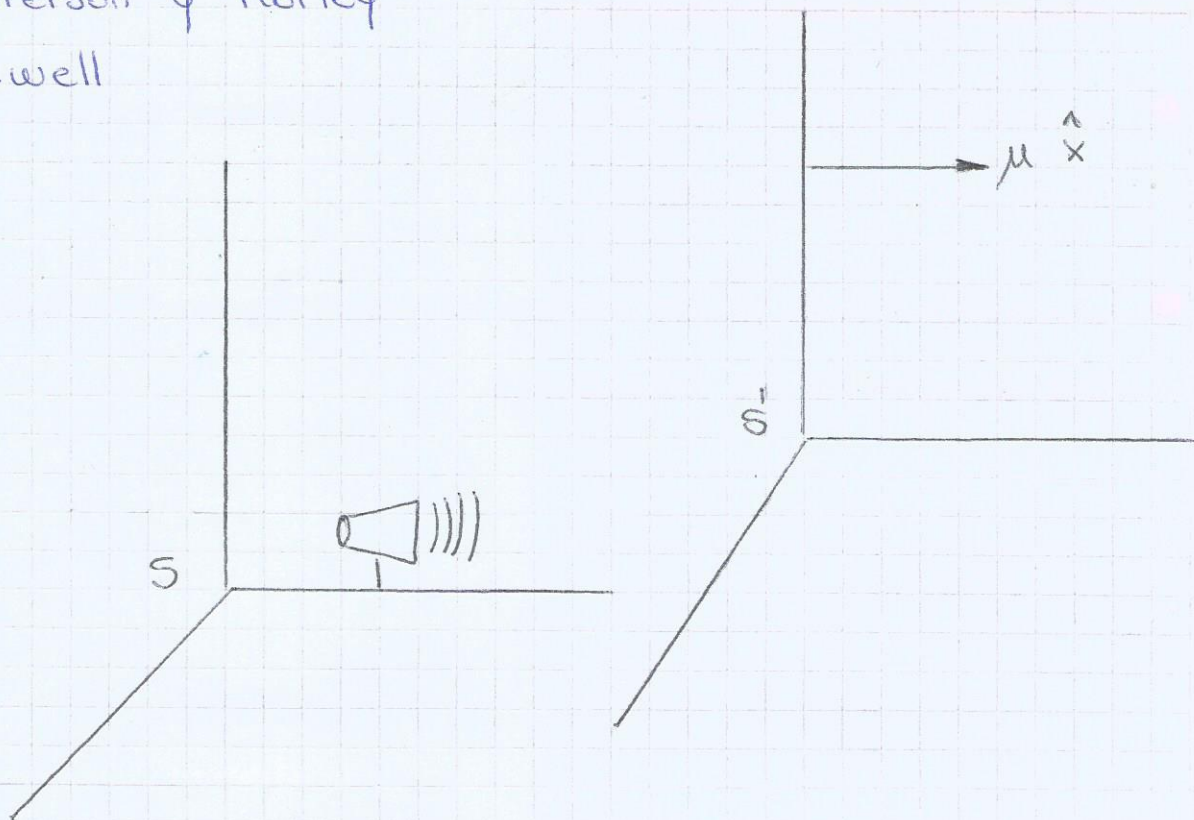
Sistema indistinguible (S.I)

$$F_{12} = -F_{21}$$

Velocidad de la luz (c) $\approx 299793 \times 10^{10} \frac{\text{cm}}{\text{seg}}$

Michelson y Morley

Maxwell



En S : hacia adelante Ns

hacia atrás $-Ns$

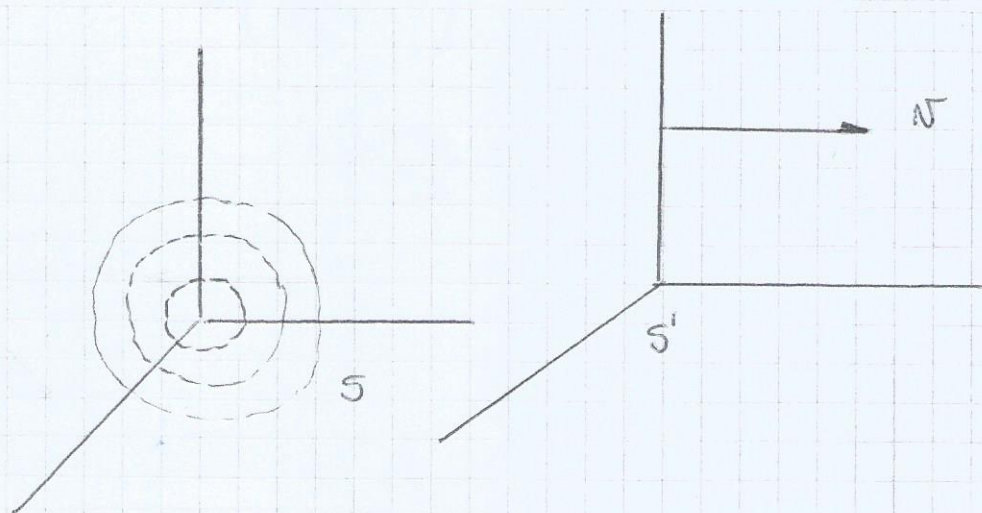
En S' : hacia adelante $Ns - u$

hacia atrás $-(Ns + u)$

Lorentz:

- Einstein (1905) "Sobre la electrodinámica de los cuerpos"
- Principio de relatividad
- La luz se propaga en el vacío con la misma velocidad en cualquier sistema de referencia

Deducción de las transformaciones de Lorentz.



$$\text{En } S: x^2 + y^2 + z^2 = c^2 t^2$$

$$\text{En } S': x'^2 + y'^2 + z'^2 = c^2 t'^2$$

• Aplicando las transformaciones de Galileo.

$$x^2 + v^2 t^2 - 2xvt + y^2 + z^2 = c^2 t^2$$

$$x' = \gamma(x - vt)$$

$$y' = y$$

$$z' = z$$

$$t' = \gamma(t + \alpha x)$$

$$\gamma^2 (x - vt)^2 + y^2 + z^2 = c^2 \gamma^2 (t + \alpha x)^2$$

$$\gamma^2 (x^2 + v^2 t^2 - 2xvt + y^2 + z^2) = c^2 \gamma^2 (t^2 + \alpha^2 x^2 + 2\alpha xt)$$

$$\gamma^2 (1 - c^2 \alpha^2) x^2 - 2\gamma^2 t x (v + c^2 \alpha) + y^2 + z^2 = c^2 \gamma^2 t^2 (1 - \frac{v^2}{c^2})$$

$$\text{Si } \alpha = -\frac{v}{c^2} \Rightarrow \gamma^2 \left(1 - \frac{v^2}{c^2}\right) = 1$$

$$x' = \gamma(x - vt)$$

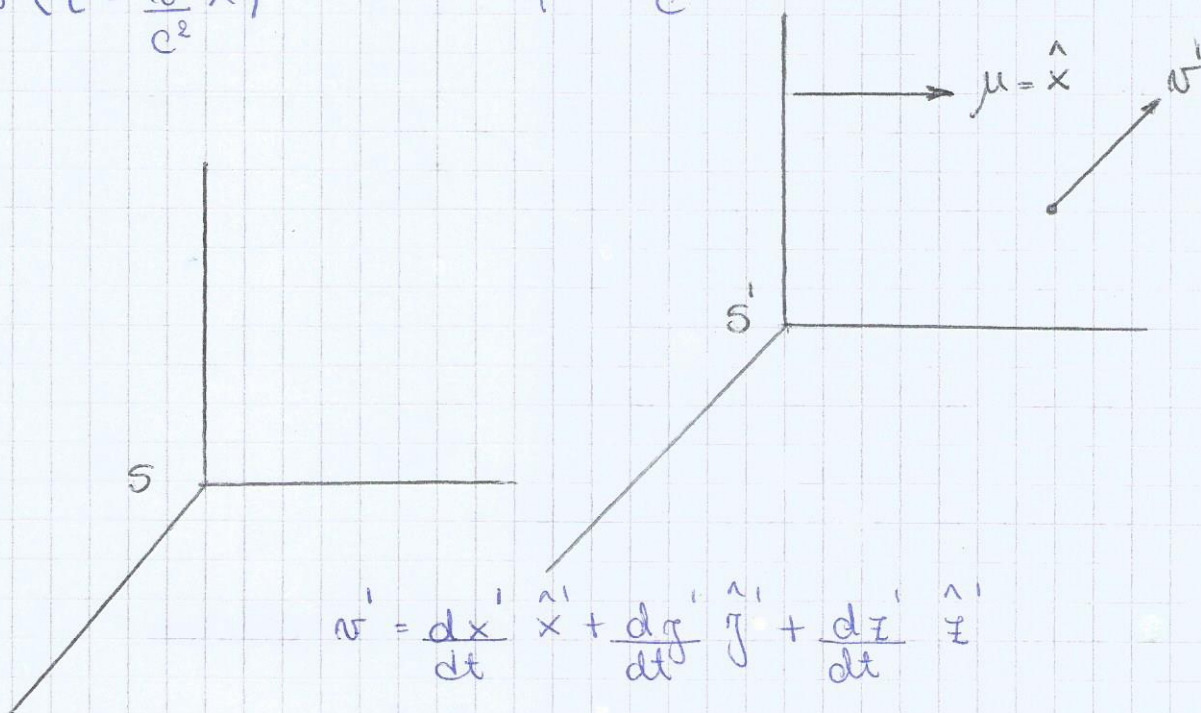
$$y' = y$$

$$z' = z$$

$$t' = \gamma(t - \frac{v}{c^2}x)$$

$$\text{con } \gamma = \frac{1}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}} > 1$$

$$\beta = \frac{v}{c} < 1$$



$$v' = \frac{dx'}{dt} \hat{x}' + \frac{dy'}{dt} \hat{y}' + \frac{dz'}{dt} \hat{z}'$$

$$\gamma = (1 - \beta^2)^{-1/2}$$

$$\beta = \frac{v}{c}$$

$$x = \gamma(x' + vt')$$

$$y = y'$$

$$z = z'$$

$$t = \gamma\left(t' + \frac{\beta}{c}x'\right)$$

$$\text{En } S: v = \frac{dx}{dt} \hat{x} + \frac{dy}{dt} \hat{y} + \frac{dz}{dt} \hat{z}$$

$$dx = \gamma(dx' + v dt')$$

$$dy = dy'$$

$$dz = dz'$$

$$dt = \gamma\left(t' + \frac{\beta}{c} dx'\right)$$

$$v_x = \frac{dx}{dt}$$

$$v_x = \frac{\gamma(dx' + v dt')}{\gamma\left(t' + \frac{\beta}{c} dx'\right)}$$

$$v_x = \frac{dx'/dt' + v}{1 + \frac{\beta}{c} \frac{dx'}{dt'}}$$

$$v_x = \frac{v'_x + u}{1 + \frac{u}{c} \frac{v'_x}{c}}$$

$$v_y = \frac{dy}{dt} = \frac{dy'}{\gamma(dt' + \frac{u}{c} dx')}$$

$$v_y = \frac{dy}{dt} = \frac{dy'/dt' \cdot \gamma}{\gamma + \gamma \frac{u}{c} \frac{v'_x}{c}}$$

$$v_z = \frac{dz}{dt} = \frac{v'_z}{\gamma(1 + \frac{u}{c} \frac{v'_x}{c})}$$

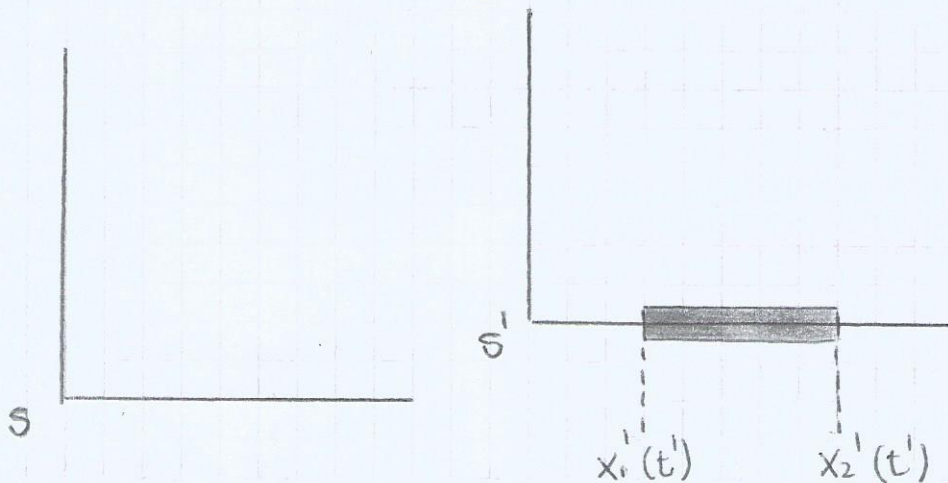
$$\text{En } S': v' = c \hat{x}'$$

$$v_x = \frac{c + u}{1 + \frac{u}{c} \frac{c}{c}} = c$$

$$v_y = 0$$

$$v_z = 0$$

1) Contracción de la longitud:



l_0 (longitud propia)

$$l_0 = x'_2(t') - x'_1(t') \text{ en } S' \quad x'_1(t')$$

$$l = x_2(t) - x_1(t) \text{ en } S \quad x_2(t)$$

Simultáneos.

$$\left. \begin{aligned} x'_2 &= \gamma(x_2 - ut) \\ x'_1 &= \gamma(x_1 - ut) \end{aligned} \right\} \begin{aligned} l_0 &= x'_2(t') - x'_1(t') \\ l_0 &= \gamma(x_2 - x_1) \end{aligned}$$

$$l_0 = \gamma l \Rightarrow L(S) = l_0(S') / \gamma$$

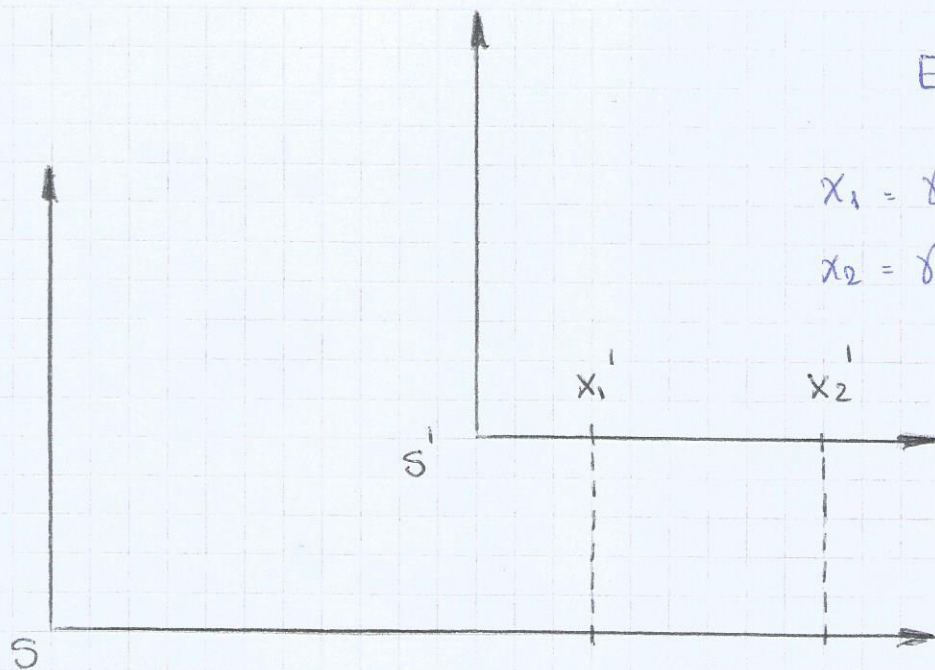
$$L(S) < l_0(S')$$

Contracción de Lorentz-Fitzgerald

En S:

$$x_1 = \gamma (x'_1 + vt'_1)$$

$$x_2 = \gamma (x'_2 + vt'_2)$$



$$x_2 - x_1 = \gamma (x'_2 - x'_1)$$

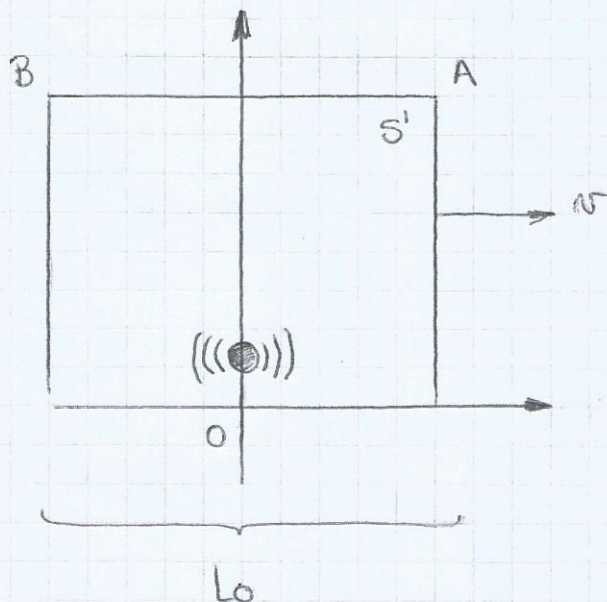
$$L = \gamma L_0$$

$$t_1 = \gamma (t'_1 + \beta/c x'_1)$$

$$t_2 = \gamma (t'_2 + \beta/c x'_2)$$

$$\Delta t = t_2 - t_1 = \gamma \frac{\beta}{c} (x'_2 - x'_1) = \gamma \frac{\beta}{c} L_0$$

2) Relatividad de la simultaneidad:



$$S' = \frac{L_0}{2} = ct'$$

$$t'_A = t'_B = \frac{L_0}{2c}$$

$$S = t_A = \gamma (t'_A + \frac{\beta}{c} x'_A)$$

$$t_A = \gamma (t'_A + \frac{\beta}{c} \frac{L_0}{2})$$

$$t_B = \gamma (t'_B + \frac{\beta}{c} x'_B)$$

$$t_B = \gamma (t'_B - \frac{\beta}{c} \frac{L_0}{2})$$

$$t_B - t_A = \gamma (t'_B - t'_A) - \frac{\beta}{c} \gamma L_0$$

$$t_B - t_A = -\frac{\beta}{c} \gamma L_0$$

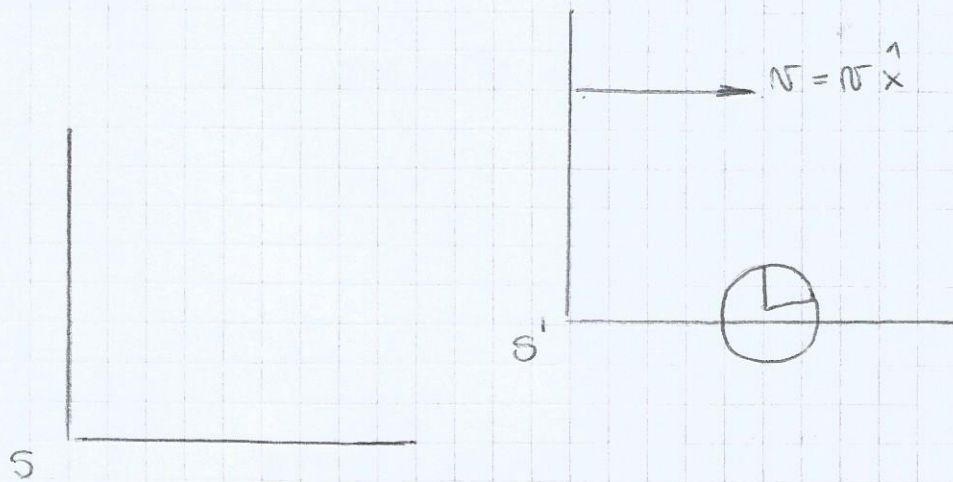
Otra forma:

$$c t_A = \frac{L}{2} + v t_A \Rightarrow t_A = \frac{L}{2(c-v)}$$

$$c t_B = \frac{L}{2} - v t_B \Rightarrow t_B = \frac{L}{2(c+v)}$$

$$t_B - t_A = \frac{-L_0 v}{\gamma (c^2 - v^2)} = -\gamma \frac{\beta}{c} L_0 \quad L = \frac{L_0}{\gamma}$$

3) Dilatación del tiempo:



Tiempo propio: $\tau = (t'_2 - t'_1)$

$$t_1 = \gamma (t'_1 + \frac{\beta}{c} x'_1)$$

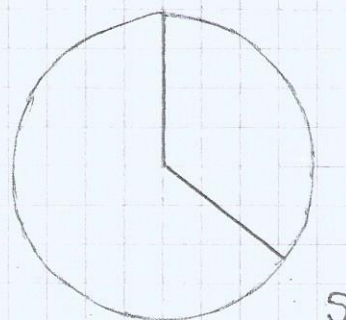
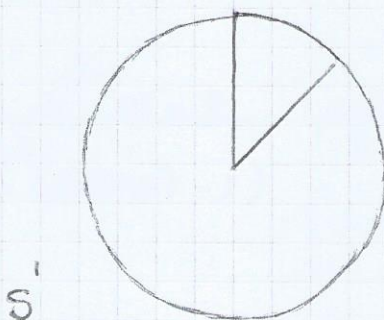
$$t_2 = \gamma (t'_2 + \frac{\beta}{c} x'_1)$$

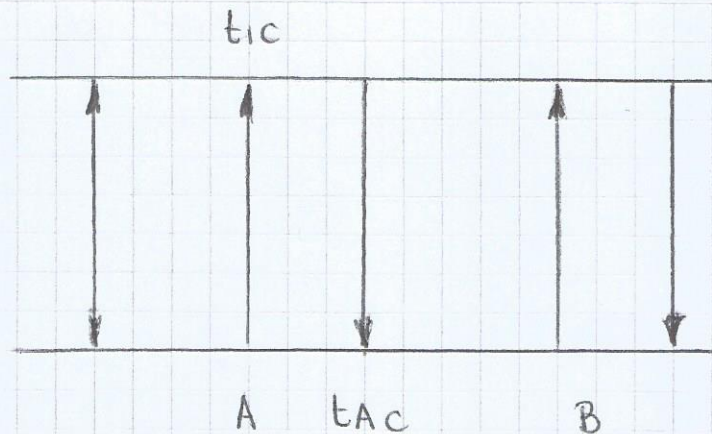
$$\Delta t = \gamma (t'_2 - t'_1)$$

$$\Delta t = \gamma \tau \text{ como } \gamma > 1$$

$$\gamma \tau > \tau$$

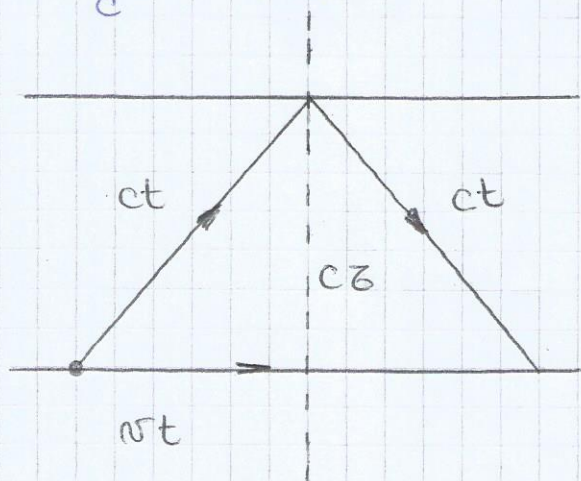
(dilatación del tiempo)





$$\vec{v} = v \hat{x}$$

$$\tau = \frac{d}{c} \text{ tiempo propio.}$$



$$c^2 t^2 = c^2 \tau^2 + v^2 t^2$$

$$t = \frac{c \tau}{(c^2 - v^2)^{1/2}} = \frac{\tau}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}}$$

$$t = \gamma \tau > \tau$$

Mesones:

Tiempo de vida: $2 \times 10^{-6} \text{ seg}$

$$\beta = \frac{v}{c} = 0,998$$

Velocidad: $299400 \frac{\text{Km}}{\text{seg}}$

$$d = v t$$

$$d = 299400 \frac{\text{Km}}{\text{seg}} \cdot 2 \times 10^{-6} \text{ seg} \quad \left. \vphantom{d = 299400 \frac{\text{Km}}{\text{seg}} \cdot 2 \times 10^{-6} \text{ seg}} \right\} \text{Relativista}$$

$$d = 600 \text{ m}$$

$$\Delta t = \gamma \tau$$

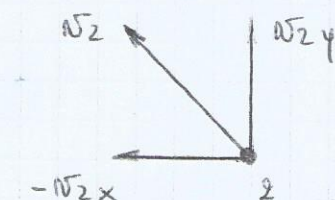
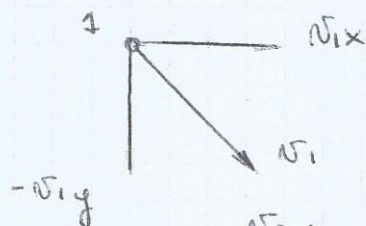
$$\Delta t = \frac{1}{\left(1 - \beta^2\right)^{1/2}} \tau$$

$$d = v \Delta t$$

$$d = 299400 \frac{\text{Km}}{\text{s}} \cdot \frac{2 \times 10^{-6} \text{ s}}{\left(1 - (0,998)^2\right)^{1/2}}$$

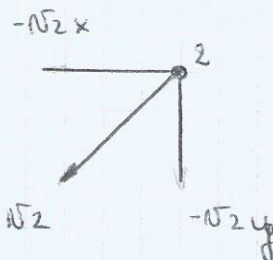
$$d = 9472 \text{ m}$$

4) Choque de partículas:



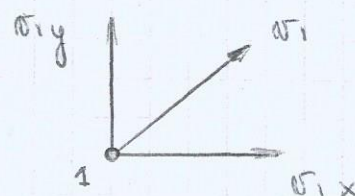
Antes

$$P_i = 0$$



Después

$$P_f = 0$$



$$m_1 = m_2 = m_0$$

$$p = m v$$

$$\Delta p_y (1) = 2 m v_y$$

$$\Delta p_y (2) = -2 m v_y$$

$$\Delta p_x (1) = 0$$

$$\Delta p_x (2) = 0$$

Supongamos que S' se mueve con velocidad $\mu = \mu \hat{x}$ respecto a S :

$$\beta = \frac{\mu}{c}$$

$$v'_x (1) = \frac{v_x - \mu}{1 + \beta v_x}$$

$$v'_x (2) = \frac{-v_x - \mu}{1 + \beta v_x}$$

$$\left. \begin{array}{l} \Delta p_x (1) = 0 \\ \Delta p_x (2) = 0 \end{array} \right\}$$

Antes del choque:

$$v'_y (1) = \frac{-v_y}{\gamma \left(1 - \beta \frac{v_x}{c} \right)}$$

$$v'_y (2) = \frac{v_y}{\gamma \left(1 + \beta \frac{v_x}{c} \right)}$$

Después del choque:

$$v'_y (1) = \frac{v_y}{\gamma \left(1 - \beta \frac{v_x}{c} \right)}$$

$$v'_y (2) = \frac{-v_y}{\gamma \left(1 + \beta \frac{v_x}{c} \right)}$$

$$\Delta p' g \neq - \Delta p' g$$

$$P_g = m_0 \frac{dg}{d\bar{t}} = m_0 \frac{dg}{dt} \frac{dt}{d\bar{t}}$$

$$d\bar{t} = \frac{dt}{\gamma_p}$$

$$\text{Con } \gamma_p = \left(1 - \frac{v^2}{c^2}\right)^{-1/2}$$

Por lo tanto :

$$P_g = m_0 \gamma_p \frac{dg}{dt}$$

$$P = \frac{m_0}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}} v$$

Cuando $v \rightarrow \infty$
 $v \rightarrow c$

$$P = m_0 \frac{dr}{d\bar{t}}$$

$$P = m_0 \frac{dr}{dt} \frac{dt}{d\bar{t}}$$

$$\frac{dt}{d\bar{t}} = \gamma > 1$$

$$P = m_0 \gamma \frac{dr}{dt}$$

$$P = \frac{m_0 v}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}}$$

Por otro lado:

$$P = m(v) \vec{v}$$

$$m(v) = \frac{m_0}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}}$$

5) Energía relativista:

$$dW = F \cdot dr$$

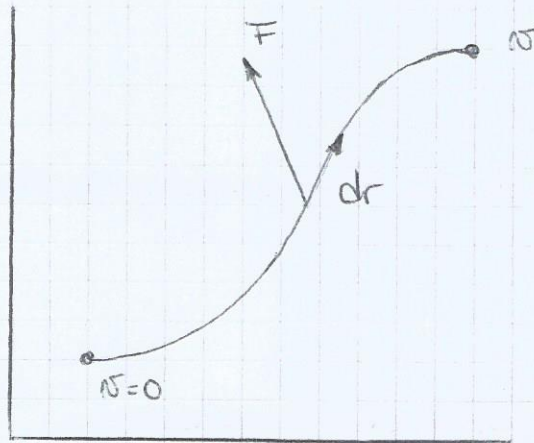
$$dW = \frac{dP}{dt} dr$$

$$dW = \frac{dP}{dt} v dt$$

$$\frac{dP}{dt} = \frac{d}{dt} \left[\frac{m_0 \cdot v}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}} \right] = \frac{m_0 \dot{v}}{(1 - \beta^2)^{1/2}} + \frac{m_0 v \dot{v} (v/c^2)}{(1 - \beta^2)^{3/2}}$$

$$v \cdot \frac{dP}{dt} = m_0 \left[\frac{v \cdot \dot{v}}{(1 - \beta^2)^{1/2}} + \frac{v v \beta^2 \dot{v} / c^2}{(1 - \beta^2)^{3/2}} \right]$$

$$v \cdot \frac{dv}{dt} = m_0 \left[\frac{v \cdot \dot{v}}{(1 - \beta^2)^{1/2}} + \frac{v^2 v \dot{v} / c^2}{(1 - \beta^2)^{3/2}} \right]$$



$$v \cdot \frac{dv}{dt} = \frac{1}{2} \frac{d}{dt} (v \cdot v) = \frac{1}{2} \frac{dv^2}{dt} = v \cdot dv = v \dot{v}$$

$$v \cdot \frac{dP}{dt} = \frac{m_0 v \dot{v}}{(1 - \beta^2)^{3/2}} = \frac{d}{dt} \left[\frac{m_0 c^2}{(1 - \beta^2)^{1/2}} \right]$$

$$dW = \frac{dP}{dt} v dt$$

$$dW = \frac{d}{dt} \left[\frac{m_0 c^2}{(1 - \beta^2)^{1/2}} \right] dt$$

$$W = \frac{m_0 c^2}{(1 - \beta^2)^{1/2}} \Big|_{v=0}^v$$

$W = E_c$ "Teorema de las fuerzas vivas"

$$W = \frac{m_0 c^2}{(1 - \beta^2)^{1/2}} - m_0 c^2 = \Delta E c$$

$$E c = \frac{m_0 c^2}{(1 - \beta^2)^{1/2}} - m_0 c^2 = m_0 c^2 (\gamma - 1)$$

Cuadrivectores:

$$S^2 = c^2 t^2 - x^2 - y^2 - z^2$$

$$\Delta S^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$$

S^2 = invariante de Lorentz

Intervalo $t_i = 0$

$x_i = 0$

Geometría pseudo eucladiana o geometría de Minkowski

$$r = (x_1, y_1, z_1)$$

$$r^L = (ct, x, y, z) \quad \text{cuadrivector posición}$$

$$i=0 \Rightarrow r^0 = ct \quad \text{con } i=0, 1, 2, 3$$

Definimos $p^i = (p_0, p)$

$$\underbrace{p_0^2 - p^2}_{\text{módulo}} = p_0^2 - p_x^2 - p_y^2 - p_z^2$$

$$p_x = m_0 \frac{dx}{dt} = m_0 \frac{dx}{dt} \frac{dt}{d\tau} = m_0 \gamma x \dot{\tau}$$

$$p_y = m_0 \frac{dy}{dt} = m_0 \frac{dy}{dt} \frac{dt}{d\tau} = m_0 \gamma y \dot{\tau}$$

$$p_z = m_0 \frac{dz}{dt} = m_0 \frac{dz}{dt} \frac{dt}{d\tau} = m_0 \gamma z \dot{\tau}$$

$$p_0 = m_0 c \frac{dt}{d\tau} = m_0 c \dot{\tau}$$

$$\begin{aligned}
 P_0^2 - P^2 &= m_0^2 c^2 \gamma^2 - m_0^2 v^2 \gamma^2 = m_0^2 \gamma^2 (c^2 - v^2) \\
 &= m_0^2 c^2 \gamma^2 \left(1 - \frac{v^2}{c^2}\right)
 \end{aligned}$$

$$P_0^2 - P^2 = m_0^2 c^2 = \text{cte}$$

$$P_0 = m_0 c \gamma$$

$$P_0 c = m_0 c^2 \gamma \Rightarrow P_0 c = E_c + m_0 c^2$$

$$E_c = \gamma m_0 c^2 - m_0 c^2, \text{ con } \gamma m_0 c^2 \text{ energía en reposo.}$$

$$E = p_0 c = E_c + m_0 c^2 \quad \text{"Energía relativista"}$$

$$p_0 c = \frac{m_0 c^2}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}} \quad c^2 = m c^2$$

$$E = c \cdot \sqrt{p^2 + m_0^2 c^2}$$

Transformaciones:

$$P_x' = \gamma \left(P_x - \beta \frac{E}{c} \right) \quad E' = \gamma (E - \beta c P_x)$$

$$P_y' = P_y$$

$$P_z' = P_z$$



$$P_1 + P_2 = 0 \quad \text{partícula producto está en reposo en SCN}$$

$$\text{En } S': P_1' + P_2' = P_3'$$

$$P_1'x + P_2'x = \gamma' \underbrace{(P_1x + P_2x)}_{0 \text{ en } S_{CM}} - \gamma' \frac{\beta}{c} (E_1 + E_2) P_3'x$$

0 en S_{CM}

$$P_3'x = \gamma' \underbrace{(P_3x - \frac{\beta}{c} E_3)}_{0 \text{ en } S_{CM}}$$

$$E_1 + E_2 = E_3$$

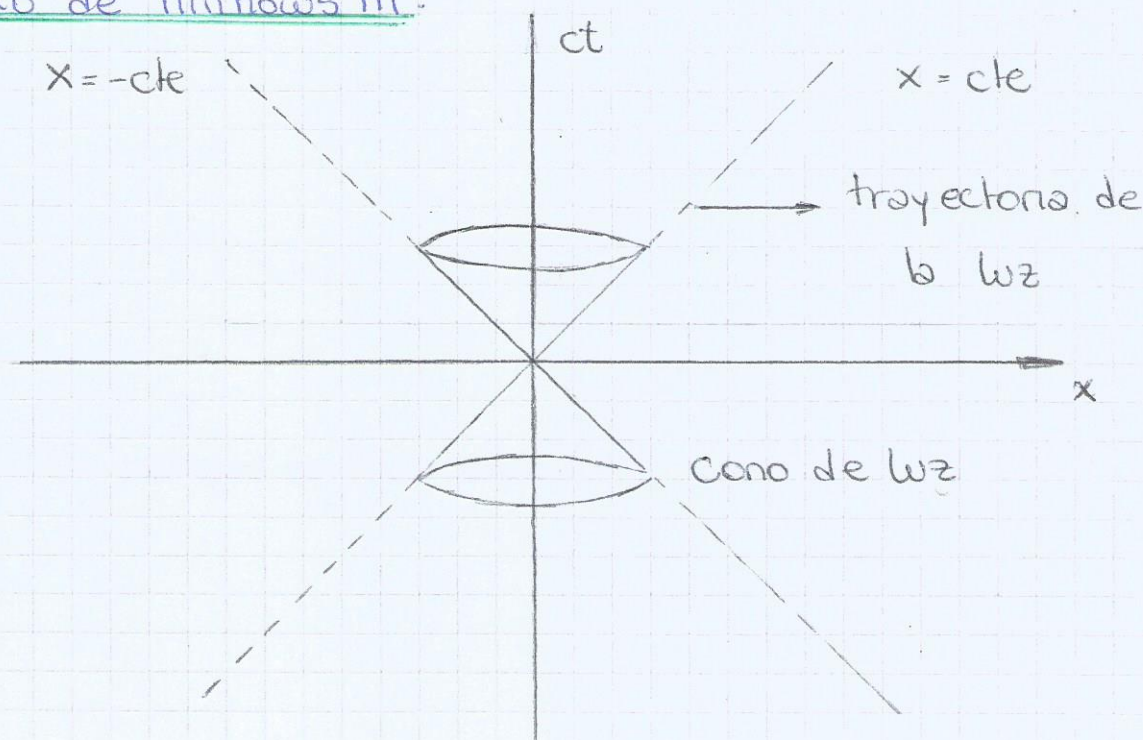
$$E_1 + E_2 = \frac{2 m_0 c^2}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}} = m_{03} c^2$$

$$m_{03} = \frac{2 m_0}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}} > 2 m_0$$

6) Leyes de Newton relativista:

$$F = \frac{dP}{dt} = m(v) \frac{dv}{dt} + \frac{dm}{dt} v \quad \neq F$$

Espacio de Minkowski:



Cada punto en este espacio se llama evento o sucesos.

Intervalo entre dos sucesos: $\Delta s^2 = c^2 t^2 - x^2$

$$\Delta s^2 > 0 \Rightarrow c^2 t^2 > x^2$$

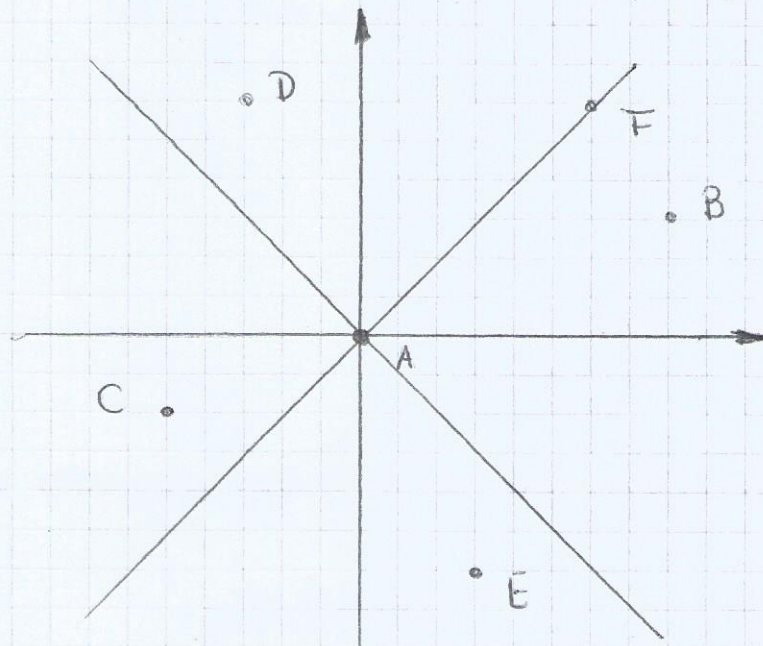
intervalo temporal

$$\Delta s^2 < 0 \Rightarrow c^2 t^2 < x^2$$

intervalo espacial

$$\Delta s^2 = 0 \Rightarrow c^2 t^2 = x^2$$

intervalo tipo luz.



Intervalo AD es tipo tiempo

Intervalo AE es tipo tiempo

Intervalo AB es tipo espacio

Intervalo AC es tipo espacio

Intervalo AF es tipo luminoso.

Supongamos que tengo un intervalo tipo tiempo, en nuestro sistema de referencia

$$c^2 (t_2 - t_1)^2 - (x_2 - x_1)^2 = c^2 (t'_2 - t'_1)^2 - (x'_2 - x'_1)^2 > 0$$

$$x'_2 = x'_1 \Rightarrow c^2 (t'_2 - t'_1)^2 > 0$$

En un tipo espacio:

$$c^2(t_2 - t_1)^2 - (x_2 - x_1)^2 = c^2(t_2' - t_1')^2 - (x_2' - x_1')^2 < 0$$

$$t_2' = t_1' \Rightarrow (\text{Simultáneos})$$

En algún sistema: $S: t_2 - t_1 > 0$

S' : en el cual $t_2' - t_1' < 0$

$$t_2' - t_1' = \gamma \left[(t_2 - t_1) - \frac{\beta}{c} (x_2 - x_1) \right] < 0$$

$$\Rightarrow \frac{c^2(t_2 - t_1)}{(x_2 - x_1)} < \gamma < c$$

$$\Rightarrow c(t_2 - t_1) < (x_2 - x_1)$$

S' se mueve con velocidad $v = v \hat{x}$ con respecto a S $w = ct$

$$x' = \frac{x - \beta w}{(1 - \beta^2)^{1/2}}$$

$$x = \frac{x' + \beta w'}{(1 - \beta^2)^{1/2}}$$

$$w' = \frac{w - \beta x}{(1 - \beta^2)^{1/2}}$$

$$w = \frac{w' + \beta x'}{(1 - \beta^2)^{1/2}}$$

$$\text{Eje } w' = x' = 0$$

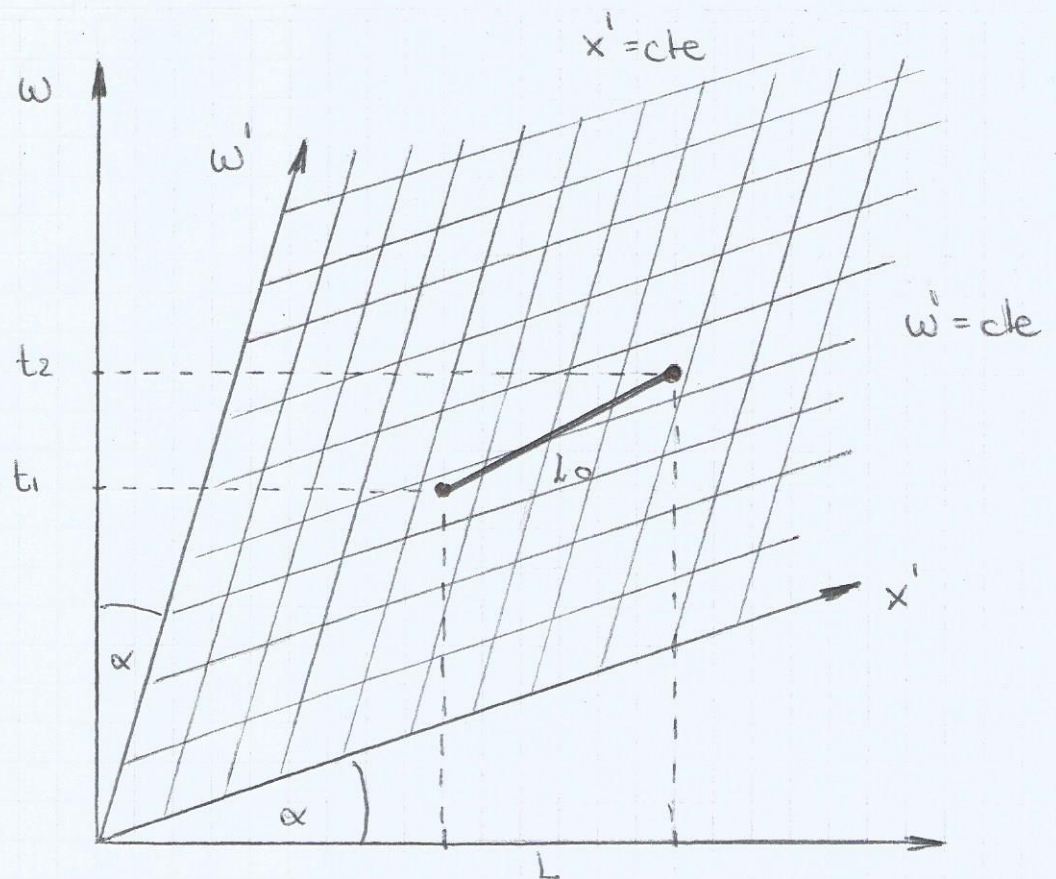
$$x = \beta w$$

$$x_2' - x_1' = l_0$$

$$\text{Eje } x' = w' = 0$$

$$w = \beta x$$

$$L = \frac{l_0}{\gamma}$$



Paradoja de los hermanos gemelos:

Δt = en Tierra

$v = 0,8 c$

τ = en nave

$\tau = 6$ años

$\Delta t = \gamma \tau > \tau$

$$\Delta t = \frac{\tau}{(1 - \beta^2)^{1/2}}$$

$$\Delta t = \frac{6 \text{ años}}{(1 - (0,8)^2)^{1/2}}$$

$\Delta t = 10$ años.